

Smart Document Assistant using Qwen, RAG and FastAPI

Abstract

Smart Document Assistant is an AI-powered application that lets users upload documents and ask questions. It combines Retrieval-Augmented Generation (RAG) with the Qwen model to generate context-aware answers.

Problem Statement

Searching long documents manually is slow and inefficient. Users need an intelligent assistant that understands document content.

Objectives

Upload documents, create embeddings, retrieve relevant content, and answer questions using Qwen.

Technology Stack

Python, FastAPI, Streamlit/React, Qwen, Sentence Transformers, ChromaDB, LangChain.

Architecture

Upload -> Text Extraction -> Chunking -> Embeddings -> ChromaDB -> Retriever -> Qwen -> Response.

Qwen Integration

Qwen receives the retrieved context from the RAG pipeline and generates grounded responses based on uploaded documents.

RAG Workflow

Document Loader, Text Splitter, Embedding Model, Vector Store, Retriever, Prompt Template, Qwen Response Generator.

Advantages

Fast, accurate, scalable, reduces hallucinations, supports multiple document types.

Future Scope

OCR, multilingual support, voice assistant, authentication, document comparison.

Conclusion

The project demonstrates an efficient document intelligence solution using Qwen and RAG.